Assignment - 1 :

① what is python and why is it called a Interpreted language.

→ python is high-level, General-purpose programming language used for web development data science, automation, artificial Intelligence, and many other application.

→ It is called an Interpreted language Because the code is executed line by line by an Interpret Instead of being converted into machine code before execution.

Eg:
python code → python Interpreted → Execution.

② what are the key features of python that make it popular for Beginners and Proterssionals.

1. Simple syntax - Easy to read & write

2. Interpreted language - No Need for complition.

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3. Dynamically typed - No Need to declare variable type.

4. Large standard - many built-in modules library

5. Cross-platform - works on windows, linux, Mac.

6. Object oriented - Support oops concepts

3. Diff b/w python 2 and python 3?

features	python 2	python 3
→ Release years	→ 2000	→ 2008
→ ~~period~~ print statement	→ print "Hello"	→ print ("Hello")
→ unicode support	→ limited	→ full unicode support
→ Integer division	→ Returns Int	→ Returns
→ Future Support	→ No longer support (end 2020)	→ Actively [illegible]

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4. what are python application in real-world project.

1. Data Science:

* Data analysis, machine learning, AI models
libraries :
pandas, numpy, Scikit-learn, Tensorflow

2. web Development:

* Backend development
* framework : Django, flask.

3. Automation:

* Automation repetitive tasks
* web security (Scraping)

4. Data Analytics:

* Data cleaning, Dashboard preparation

5. cyber Security:
* Penetration testing tools.

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what is PEP 8 and why is it important in python programming:

→ PEP 8 stands for python Enhancement proposal 8

→ It is the Aficial style guide for writing python code:

↳ It defines Rules like:
* Indentation (4 spaces)
* variables Naming Conventions
* maximum line length (79 char)
* proper code formatting

Eg: x=10
y=20
print(x+y)

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6. who developed for python and in which year was it Released?

→ python was created by Guido Van Rossum

→ First Released : February 20, 1991
Developed at Centrum wiskunde & Informatica (CWI) : Netherland.

7. what does "dynamically typed" in python?

→ In python you do not declare the data type of a variable

→ The interpreter automatically determines the type during execution.

Ex : x = 10
y = "Hello"

→ The same variable can store different data type

8). Diff b/w compiler and Interpreter.

Compiler	Interpreter.
→ Converts entire program into Machine code	→ Excete code line by line
→ Faster execution	→ slower execution
→ Error shown after compilation	→ Error shown immediatily
→ Example. C, C++	→ Example : python.

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